# BIOMETRIC BLOOD GROUP IDENTIFICATION SYSTEM

Final Year B.E. Major Project • Deep Learning AI Analysis Report

**MEDICAL DISCLAIMER:** AI-based preliminary estimation only. This result must be confirmed using a standard laboratory blood typing test before medical use. This system does not replace standard serological blood typing.

## 1. EXAMINATION & SAMPLE AUDIT DETAILS

| | | | |
|---|---|---|---|
| **Report ID:** | BG-000340 | **Timestamp:** | 2026-09-06 19:12:04 |
| **Input Source:** | Mantra MFS100 Scanner \| Standard Finger: LEFT THUMB | **Evaluation Status:** | LOW_CONFIDENCE |
| **Model Name:** | Champion Ensemble V2 (ResNet50 58% + VGG16 38% + EffNet 4%) | **Confidence:** | **39.68%** |
| **Standard Acquisition Finger:** | **LEFT THUMB** (Convention-based protocol — not automatically verified by scanner) | | |

## 2. BIOMETRIC IMPRESSION & PREDICTED BLOOD GROUP



**Predicted Blood Group:**

# AB+

Confidence Score: **39.68%**

Threshold: 60.0%

Classification: Low Confidence - Retest Required

Derived ABO Group: **AB** (48.5% prob) | Rh Factor: **+**

**Rh FACTOR UNCERTAINTY ADVISORY:** Rh factor (±) determination is uncertain (moderate 8-class confidence (39.7%)). ABO group appears to be 'AB' (48.5% confidence). Standard serological laboratory blood typing is required to confirm Rh factor.

## 3. 8-CLASS BLOOD GROUP PROBABILITY DISTRIBUTION

| A+ | A- | AB+ | AB- | B+ | B- | O+ | O- |
|---|---|---|---|---|---|---|---|
| 4.7% | 1.4% | 39.7% | 8.8% | 22.0% | 2.9% | 1.9% | 18.7% |